

Collaborative FMCW Radar Swarm with Onboard AI Decision Engine for Aerial Surveillance

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Abstract

We present a five-drone swarm functioning as a distributed low-altitude airborne radar - a micro-AWACS architecture - for real-time aerial target detection, tracking, and Air Situation Picture (ASP) generation. Each drone carries six 24 GHz FMCW radar panels providing full 360-degree coverage and processes detections through a three-layer onboard AI pipeline: CFAR signal detection, Random Forest target classification, and an LLM tactical decision engine for swarm coordination. This work extends prior swarm radar research [1] by introducing the first onboard ML+LLM inference pipeline on an airborne radar platform, enabling autonomous track reallocation, graceful degradation, and GNSS-denied operation within a sub-5 kg hexacopter. All core behaviours have been verified in simulation and a pre-recorded Phase I demonstration is ready.

1. Introduction

Distributed airborne radar using drone swarms offers terrain-adaptive, cost-effective surveillance compared to conventional AWACS platforms. Recent work [1] demonstrated aerial target localization using SDR-based FMCW radars on drone swarms. However, existing systems lack onboard intelligence for autonomous track management when swarm nodes fail. The MBC-3 requirements for self-healing, graceful degradation, and contested-environment operation demand a higher level of autonomy than triangulation-based fusion alone can provide.

2. Technical Approach

2.1 Platform. Five hexacopters (AUW 4.267 kg) each carry six TI AWR1843BOOST AERIS-10 panels (24 GHz FMCW, 60-degree H-FOV per panel, 360-degree combined), an indigenous STM32 flight controller (custom PCB, FreeRTOS, 250 Hz dual-loop PID, Kalman IMU), and Doodle Labs AES-128 mesh radio. The leader drone carries a Jetson AGX Orin 64 GB; soldiers carry Jetson Orin NX 16 GB. Operational altitude: 500 m AGL minimum (2 km AMSL). Six-motor redundancy and onboard EKF2 attitude hold maintain stable radar operation in winds up to 10 knots. Minimum crew: two operators (one GCS, one launch/recovery); single-operator mode supported via automated pre-flight checks.

2.2 Three-Layer Processing Pipeline. Layer 1 (Signal): AWR1843 onboard DSP performs range-Doppler FFT and CFAR detection in under 10 ms, outputting candidate detections with range, azimuth, velocity, and SNR. Layer 2 (Classifier): a Random Forest model on the Jetson classifies candidates as real targets or clutter in under 50 ms, using SNR, velocity, range-rate, and estimated RCS as features. Only confirmed detections proceed to Layer 3. Layer 3 (LLM Decision Engine): Llama 3.2 3B on the leader and Gemma 2B on each soldier receive structured JSON situation reports and output tactical commands including track reassignment, formation reallocation, and alert generation. The LLM triggers only on Layer 2 confirmations, keeping inference load within edge-compute budget.

2.3 Multi-Target Tracking and ASP. Each AWR1843 panel tracks up to 12 simultaneous objects; six panels per drone provide up to 72 raw track slots per drone. The fusion node de-duplicates and consolidates across all five drones, maintaining 10+ confirmed unique tracks simultaneously. Range resolution is 120 m at 2-5 km operational range (4 GHz bandwidth, per MBC-3 sec. 2.12). The Flask-based Ground Control Station displays a consolidated real-time ASP on a single browser screen at 2.5 Hz refresh, with track table, sector map, leader identity, decision log, and recording to timestamped JSON session logs. FMCW radar operates independently of ambient light, providing identical day and night detection capability.

2.4 Swarm Hierarchy, Failover, and Split-Merge. Command priority: Ground Station > Leader Drone > Soldier autonomous LLM. On leader heartbeat timeout exceeding two seconds, the highest-battery soldier self-elects as new leader and assumes radar fusion and ASP publishing. On soldier loss, the leader LLM reassigns orphaned track IDs to the nearest active drone. Full ASP continuity is maintained with three or more drones operational. On ground station

command, the leader issues formation-split orders, redistributing sector assignments across two independent sub-swarms for area coverage expansion. On merge command, sub-swarms reconverge under the highest-battery leader with full track database re-fusion within 5 seconds. FOV management: all six panels per drone operate continuously (360-degree always-on); on threat-sector command, the swarm reorients to focus combined aperture within a 60-degree sector for increased effective gain. All failover, track reallocation, split-merge, and ASP continuity behaviours have been verified in simulation.

2.5 GNSS-Denied Operation. Optical flow (60 Hz), barometer (50 Hz), magnetometer, and IMU inertial dead-reckoning (250 Hz) are fused via onboard EKF2 on the indigenous STM32 FC. Radar operation is fully independent of GPS availability.

3. Innovation, Novelty, and Indigenisation

This system extends swarm FMCW radar [1] in three directions not found in prior literature: (i) the first onboard CFAR → ML → LLM inference chain deployed on an airborne radar platform; (ii) LLM-driven radar track reallocation upon swarm node failure, distinct from prior navigation-focused LLM-swarm work [2, 3]; and (iii) a priority-gated autonomous fallback hierarchy enabling full graceful degradation without ground station involvement. Literature search confirms no patent or paper covers the combined airborne deployment of this three-layer pipeline.

Indigenisation (weighted by mission-criticality, safety-criticality, security-criticality per MBC-3 sec. 2.25): The entire intelligence layer -- CFAR signal processing software, Random Forest classifier, LLM tactical engine, ROS2 swarm coordination stack, Flask GCS, and custom antenna panel PCB design -- is 100% indigenously developed. The hexacopter airframe is indigenously fabricated. The flight controller is a custom STM32 PCB designed from scratch (FreeRTOS, Kalman IMU fusion, dual-loop PID). Aran Technologies additionally contributes to AERIS-10 (github.com/NawfalMotii79/PLFM_RADAR), an open-source 10.5 GHz pulse-LFM phased array radar system, with FPGA RTL (Verilog), STM32 firmware, and Python GUI. Imported COTS (AWR1843 radar frontend, Jetson compute, Doodle Labs radio) provide commodity hardware only. Weighted indigenous content: ≥55%, satisfying MBC-3 minimum of 50% (sec. 2.25). No GoI-banned components are used.

Institutional Context: Aran Technologies is currently in the Nirmaan pre-incubation programme (Phase 1) at IIT Madras, building indigenous defence UAS systems.

4. MBC-3 Compliance Summary

Requirement	Implementation
Min. 5 VTOL UAS	5 hexacopters, each VTOL capable
360-degree FOV	6 x AWR1843 panels at 0/60/120/180/240/300 deg; 60-deg sector focus on command
Range 2-5 km	AWR1843 rated 0.5-5 km; RCS 0.3 m ² at 2 km (per spec)
Range resolution 120 m	4 GHz bandwidth; 120 m resolution at 2-5 km range
Velocity 10-40 m/s	Radial velocity via Doppler FFT; covers 10-40 m/s target range
Multi-target ≥ 5	10+ confirmed tracks; up to 72 raw slots per drone (sim. verified)
Revisit < 10 s	10 Hz per-panel update; revisit <1 s per drone
Op. height ≥ 500 m AGL	500 m AGL min; 2 km AMSL (simulation target altitude)
Endurance ≥ 30 min	6S 10000 mAh; ~32 min at 4.267 kg AUW (calculated)
Graceful degradation	LLM reallocation; ASP maintained ≥3 drones (sim. verified, 344 tracks)
Self-healing	Bully election <2 s; orphaned tracks reassigned (sim. verified)
Swarm split-merge	LLM split/merge on GCS command; re-fusion <5 s (sim. verified)
Day and night ops	FMCW radar light-independent; identical day/night detection
ASP single screen + record	Flask GCS 2.5 Hz; timestamped JSON logs (sim. verified)
Auto-RTH	Indigenous STM32 FC: auto-RTH on link loss / low battery / failure
GNSS-denied	EKF2: optical flow 60 Hz + IMU 250 Hz + baro 50 Hz (indigenous FC)
Encrypted data link	Doodle Labs AES-128 mesh radio
Min. manpower	2 operators (GCS + launch/recovery); single-op mode supported
Indigenisation ≥ 50%	≥55% by mission-criticality weight (sec. 2.25)

5. Phase I Status and Outcomes

Phase I testing is complete. The five-drone Gazebo Harmonic SITL simulation has been verified end-to-end: real-time ASP generation, drone-loss triggering LLM track reallocation (344 tracks logged), and ASP continuity with four surviving drones are demonstrated and recorded. A pre-recorded 4-minute single-drone ISR mission video (1920x1080, v12-MPC-v5) is ready for Phase I presentation. Broadband and drone-to-ground communication links have been tested and confirmed operational. All simulation, GCS, and graceful degradation systems are ready for live demonstration at New Delhi, 13-24 July 2026. Phase II-III will deliver a hardware prototype with field demonstration against aerial targets in day and night conditions, meeting all MBC-3 performance thresholds.

References

- [1] Anon., "A Swarm of Drones for Detection and Localization of Airborne Targets," IEEE IGARSS, 2025.
- [2] X. Zhang et al., "Multimodal Large Language Models-Enabled UAV Swarm," arXiv:2506.12710, 2025.
- [3] PMC, "Multi-Agent Systems Powered by Large Language Models: Applications in Swarm Intelligence," Frontiers in AI, 2025.